

MATH 1342 – Elementary Statistics – Week 15: ANOVA and Putting It All Together

Dec 1 – 3 | 8.1 Analysis of Variance • Review: Putting It All Together – From Sampling to Hypothesis Testing | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 7. Apply ideas of appropriate sampling techniques and experimental design to data production. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations. 11. Demonstrate skills in hypothesis testing using chi-squared tests. 12. Demonstrate skills in inference for regression and ANOVA techniques.	1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CO12) 2. Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0 . (CO12) 3. Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CO4, CO12) 4. Choose the correct test among one- and two-proportion z, one- and two-sample t, paired t, and chi-square, and carry out a complete six-step hypothesis test. (CO10, CO11) 5. Explain what the study design (random sampling vs. random assignment) allows a conclusion to claim, and distinguish statistical from practical significance. (CO7, CO10)	e-Text: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	MO1, MO2, MO3	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 8.1 Analysis of Variance	MO1, MO2, MO3				
		Slides: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	MO4, MO5		
		Review Slides: Putting It All Together (with answer key and no-answers versions)	MO4, MO5				
		StatKey – Theoretical Distributions (F)	MO3				